TransitPro · Transport Management System

App Overview · RDBMS Project

A complete web application that manages a logistics company's **shipments, fleet, drivers, routes, payments and maintenance**, built on a normalised **MySQL** database designed from an **ERD**.

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7	6	19	73	9	1
TABLES	FOREIGN KEYS	CHECK RULES	SAMPLE RECORDS	SQL REPORTS	CLICK TO RUN

How it works

Browser — Dashboard, forms, search, reports

⇄

Flask (Python) — Validation, rules, SQL via PyMySQL

⇄

MySQL 8 — transport_db, 7 tables · 1 view

Run it in 3 steps

1. **Unzip the project folder**
   Everything needed is inside, including the SQL scripts and ERD.
2. **Double-click `START_HERE.bat`**
   Installs packages automatically and asks for the MySQL root password.
3. **Use the app at `127.0.0.1:5000`**
   The database **transport_db** is created in MySQL on the first run. A green **MySQL connected** badge confirms the link.

What you need

- **Python 3.10+** — Free from python.org, with "Add to PATH" ticked
- **OPTIONAL** — **MySQL 8 + Workbench** — Without it, the app uses a built-in SQLite copy
- **Any browser** — Edge, Chrome or Firefox. Works on mobile too.

SQL concepts demonstrated

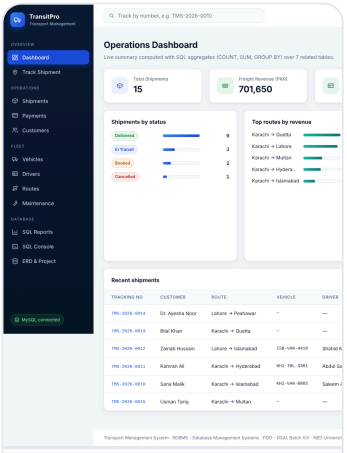
Primary Keys · Foreign Keys · 3NF · NOT NULL · UNIQUE · CHECK · DEFAULT · INSERT · SELECT · UPDATE · DELETE · INNER JOIN · LEFT JOIN · RIGHT JOIN · SELF JOIN · GROUP BY / HAVING · Sub-queries · View · Index · Stored Procedure · Function · Triggers · COMMIT / ROLLBACK · GRANT / REVOKE

Key features

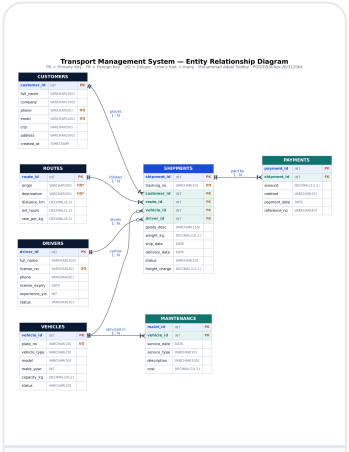
- **Live dashboard** — Revenue, balances and shipments by status and route
- **Full CRUD** — Add, view, edit and delete for all 7 tables
- **Search & filter** — Keyword search on every list, status filters
- **Shipment tracking** — Find any shipment by tracking number with a progress bar
- **SQL Reports** — 9 live queries, shown with their SQL text
- **SQL Console** — Run your own SELECT queries in the browser

What makes it stand out

- **One-click setup:** the database builds itself in MySQL. No commands to type.
- **SQL made visible:** every save shows the exact statement that ran.
- **The database guards the data:** constraint violations are caught and explained clearly.
- **Real business rules:** automatic freight (weight × rate), balance due, capacity warning.
- **Secure & safe:** parameterised queries prevent SQL injection, and every save runs in a transaction.
- **Professional, responsive UI:** works on laptop, tablet and phone.



Operations dashboard



ERD: 7 entities, 6 relationships